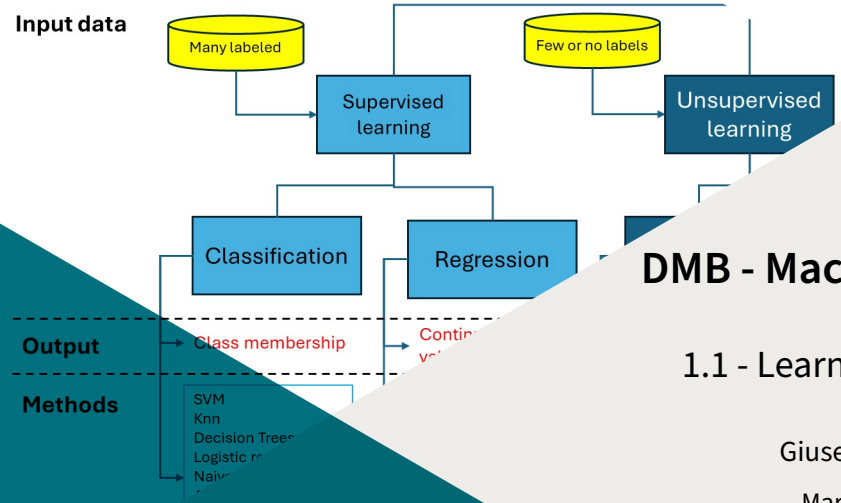

DMB - Machine Learning

1.1 - Learning Paradigms

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Summary

1. Learning paradigms: supervised, unsupervised, Reinforcement, ecc.

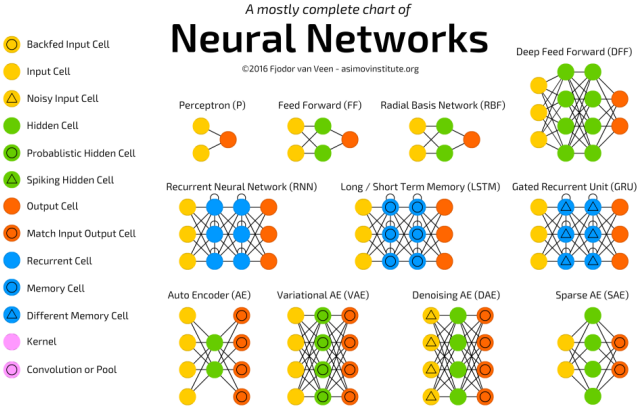
2. Supervised Learning

3. Unsupervised learning

4. Reinforcement learning

5. Some resources

- Ada Lovelace (1843) ... *the analytical engin has no pretentions whatever to originate anything. It can do whatever we know how to order it to perform.... Its province is to assist us in making available what we're already acquainted with.*

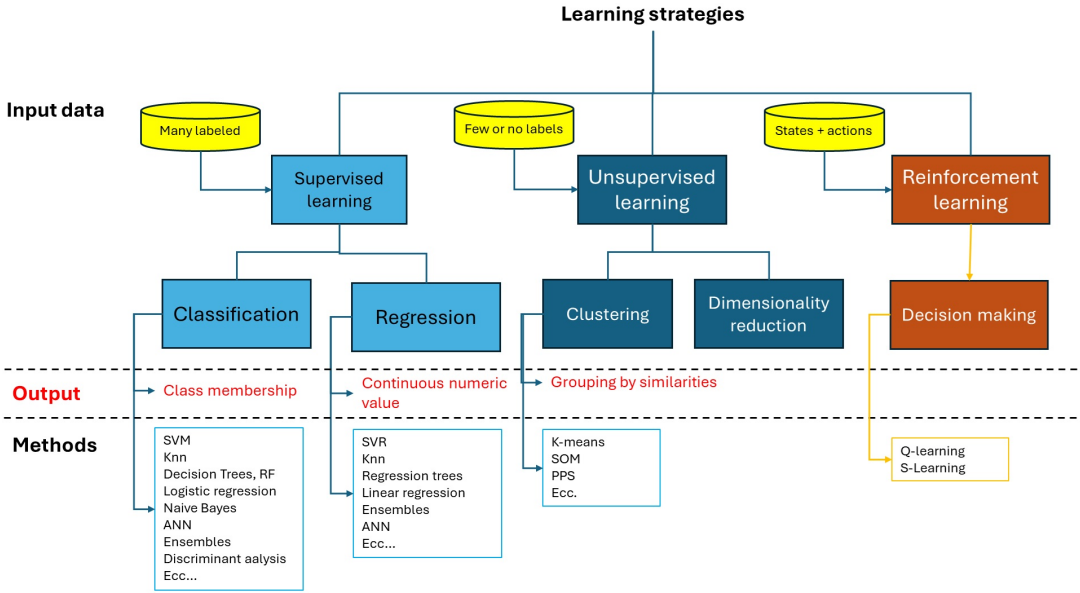


(©G.Longo).

A ML model learns how to transform (rules) input data into meaningful outputs by being exposed to known examples of **input-output** pairs.

Therefore ML methods learn how to transform data into useful output.

Learning paradigms: supervised, unsupervised, Reinforcement, ecc.


Learning strategies
Input data
Many labeled
Few or no labels
States + actions
Supervised learning
Unsupervised learning
Reinforcement learning
Classification
Regression
Clustering
Dimensionality reduction
Decision making
Output
Class membership
Continuous numeric value
Grouping by similarities
Methods
SVM
Knn
Decision Trees, RF
Logistic regression
Naive Bayes
ANN
Ensembles
Discriminant aalysis
Ecc...
SVR
Knn
Regression trees
Linear regression
Ensembles
ANN
Ecc...
K-means
SOM
PPS
Ecc.
Q-learning
S-Learning

(©G.Longo).

In the past, Machine Learning methods were usually split in two broad catagories (or Learning Paradigms) according to the kind of **Experience** (E) they are allowed to have during training: **supervised** or **unsupervised**.

- **Supervised**. The method learns by examples. In other words, the input data have a predefined answer to the task at hand (called **target**) and the method uses the target to learn the underlying mapping function.

- **Unsupervised**: the method learns from the input data themselves the probability distribution that generated the data set either implicitly (e.g. as in denoising) or explicitly (e.g. as in density estimation). A special class of unsupervised methods deals with **clustering**, i.e. in partitioning the input space in homogeneous groups (or clusters).

We wish to stress that, contrary to the past, the distinction between supervised and unsupervised is no longer that important since the boundary between the two approaches has become quite blurred.

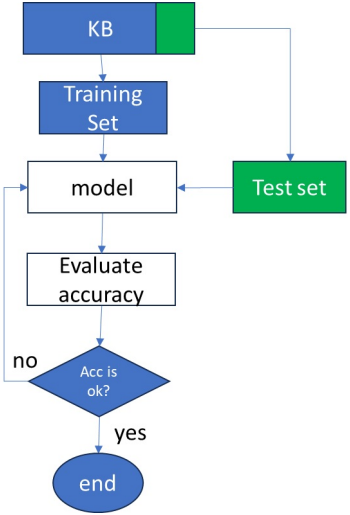
Supervised Learning

Usually large sets of **labeled** (i.e. data for which you already know the desired output) are needed. We shall refer to these data as **Knowledge Base**.

The model uses the a-priori knowledge contained in a subset of the KB, the **Training Data Set**, to learn the classification rule.

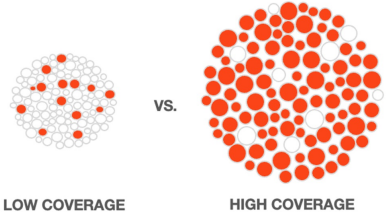
The remaining data in the KB (**Test Data Set**) are used to evaluate the accuracy of the model. If accuracy is good the model is *frozen*.

The frozen model is then applied to the data for which we want to know the answer.



It is impossible to emphasize enough the importance of EDA in a supervised task !!!

- Labeled data are rare and often very noisy. They may be biased, incomplete, ecc.
- The training set should sample in a homogeneous way the wholre region of the OPS covered by the data
- the density of training points in the OPS decreases if we increase the number of features used (Curse of dimensionality).
- The quality of the results depends on the quality of the data... remember the famous phrase: **... garbage in, garbage out** (also known as the **GIGO rule**).



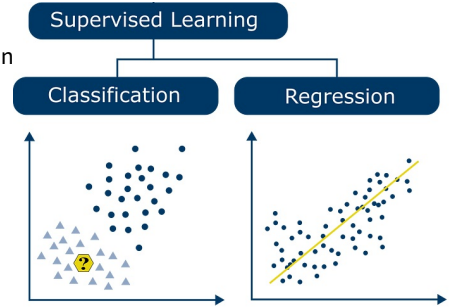
Esemplification: red dots are training data grey circles are the whole data. (©G.Longo).

Garbage in
Garbage OUT

Main supervised Tasks

Supervised Learning is mainly used for classification or regression problems.

- **Classification**:(output is discrete or categorical)
- **Regression**: (output is continuous)



What is a classification?

Given a set $\mathbb{S}$ composed of n elements x_j, an ideal classification is a partition of $\mathbb{S}$ in m subsets $\mathbb{S}_i$ (where $m << n$) such as:

- $\forall x_j$ in $\mathbb{S}, \exists! \, \mathbb{S}_i : \; x_j \in \mathbb{S}_i$;
- $\forall (i,j), \; i \neq j, \; \mathbb{S}_i \cap \mathbb{S}_j = \emptyset$;
- $\cup_{i=1}^{m} \mathbb{S}_i = \mathbb{S}$

The above is called a **Crispy Classification**.

If the first condition is violated, then an object may belong (with different levels of likelyhood) to more than one class and we have a **Fuzzy Classification**.

In the language of ML, the computer needs to learn how to place an input vector $\mathbf{x} \in \mathbb{R}^D$ in one among K different classes (C_i ß $= 1, \ldots, K$ (in other words, to uncover the mapping function $f : \mathbb{R}^D \to \{1, 2, \ldots, K\}$), where D is the number of input features.

In Crispy Classification, the OPS is divided into different **decision regions**, separated by **decision boundaries** or **decision hypersurfaces**.

Examples:

- Image classification
- Character recognition
- Customers targeting

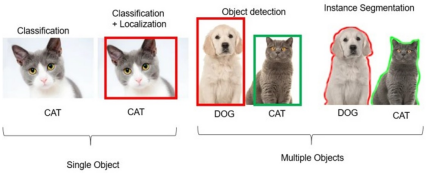


Figure 1: Some classifications(©Common Licence).

Classification becomes more difficult if the input data are incomplete, i.e. if some features are not measured or not available for some of the input examples.

In this case, the machine needs to learn not just one function f, but a set of functions $f_1, f_2, \ldots, f_n$, each specialized in predicting the output when a given combination of input features is missing.

Examples: Classification with incomplete data is a very common problem in medical diagnosis where in the input vector (which may include as features the results of some specific exam or test) some features are very often missing.

We shall see how to deal with this type of problems later on.

Regression

- **Regression**. In this case the task is learning the function $f : \mathbb{R}^n \to \mathbb{R}$ mapping the input examples to a numerical value.

Examples: evaluation of insurances claims, algorithmic trading, distances of galaxies, etc.

Unsupervised learning

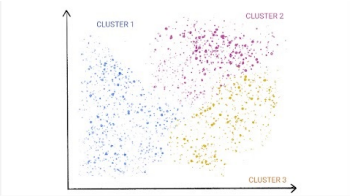


Figure 1. An ML model clustering similar data points.

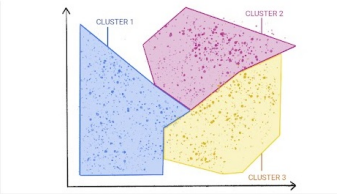


Figure 2. Groups of clusters with natural demarcations.

Example of clustering (©Common Licence).

Unsupervised methods

Unsupervised learning algorithms are better suited for more complex processing tasks, such as organizing large datasets into clusters of similar data.

They are useful for identifying previously undetected patterns in data and can help identify features useful for categorizing data.

To understand the patterns you need to go through the data groupings and attempt to classify them based on your understanding of the dataset, such as:

Limitations

- It is often difficult to understand the meaning of a given set of clusters. Different approaches are possible:
  - using a small subset of labeled data;
  - studying the characteristics of the clusters (parameter distribution, etc)....
- different clustering algorithms usually lead to different sets of clusters. It is difficult to evaluate which model ise better than the other.

Reinforcement learning

Structured output tasks

- **Parsing**. mapping an input to a structure (usually a tree) which describes its hierarchy of concepts.

Examples: mapping a Natural language sentence to its structure in terms of verbs, nouns, adverbs, etc. or assigning each pixel of an image to a specific class, automatic image captioning.

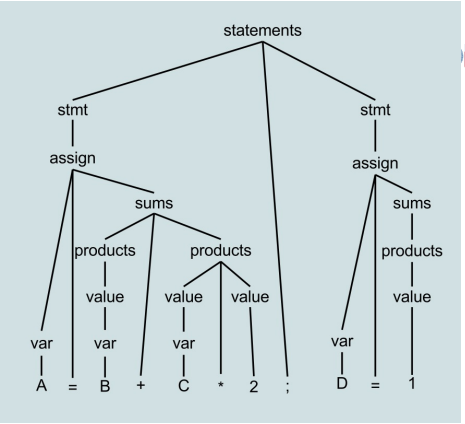


Figure 2: Some classifications(©Common Licence).

Transcription

Following [GBC, p. 98], *... the machine is asked to observe a relatively unstructured representation of some kind of data and transcribe the information into discrete textual form*.

Examples: optical character recognition, speech recognition, etc.

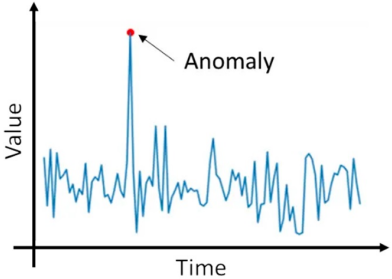
Machine translation

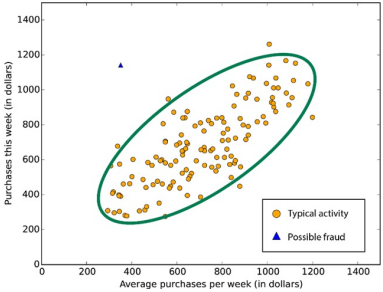
Examples: converting speech or text from one language to the other.

Anomaly detection

Identifying unusual inputs or unusual events:

- Aberrant points in time series;
- Data points which lay far away from all clusters;
- Objects which cannot be attributed to one class only (fuzzy classification);
- Etc.





VERY Important Warning

Most of ML developments are application and profit driven.... and this is terribly dangerous.

You can easily download a software from github, apply it to your problem and obtain results which almost always seem good **... but** 95% **of the times are not good, unreliable, non optimal and dangerous.**

The only antidote is understand deeply the mathematics and the statistics behind the model and perform a robust evaluation of the results.

Some resources

Homeworks

First

Form the groups (2 people. Write the group's composition using the dedicated Teams channel.

-

Second

Identify in your every day life which tasks could be addressed as classfication, which as clustering, which as regression.